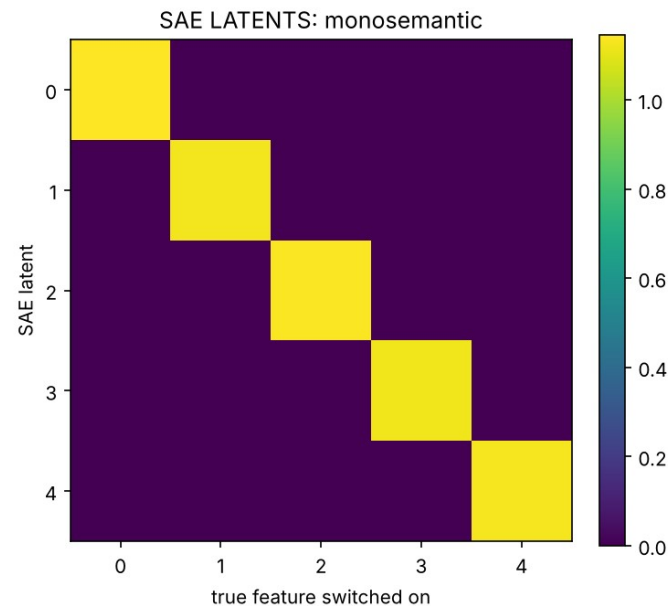
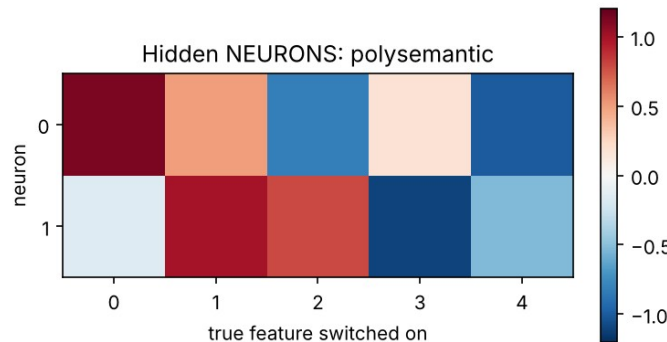
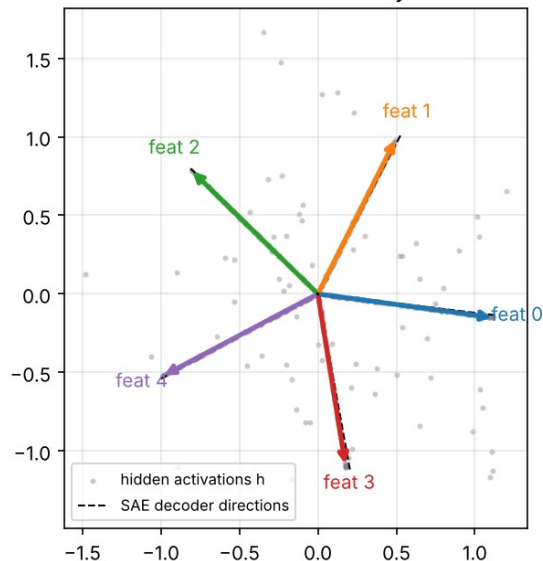


5 features in a 2-D space (superposition)
dashed = directions found by the SAE



Superposition

Models represent more features than they have dimensions, as almost-orthogonal directions. It works because features are sparse: rarely active together.

Polysemantic neurons

So a single neuron mixes several features: reading neurons one by one gives a confusing picture.

Sparse autoencoder

Reconstruct the model's activation using an encoder-decoder network.